

Noise and Outliers

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can calculate mean and median.

I can investigate an unusual value before deleting it.

I can explain how a summary can hide variation.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Overfitting Intuition. Give an example and a mistake that the idea helps you avoid.

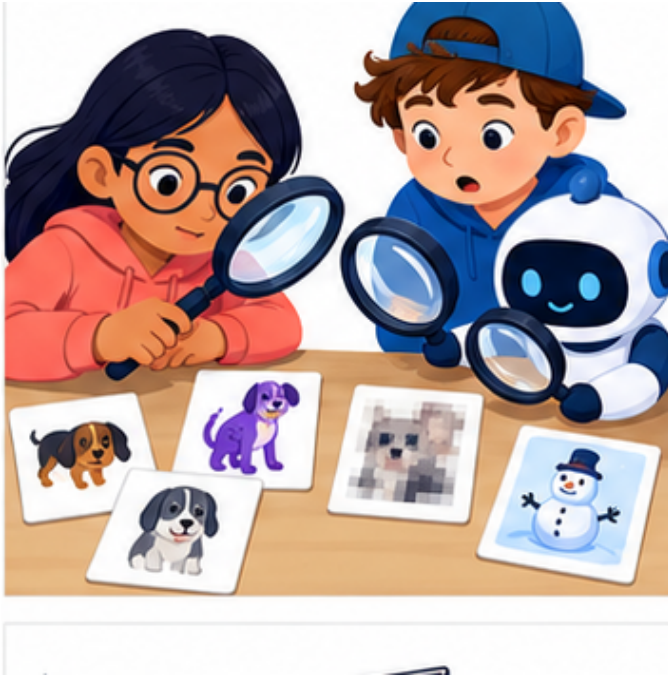
Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Training and Testing. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 17: Noise and Outliers

Mistakes and unusual values should be investigated, not automatically erased.



THINKER MISSION

Decide whether four odd records are errors or real cases.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

What is the total of 2, 3 and 4?

This week's end goal

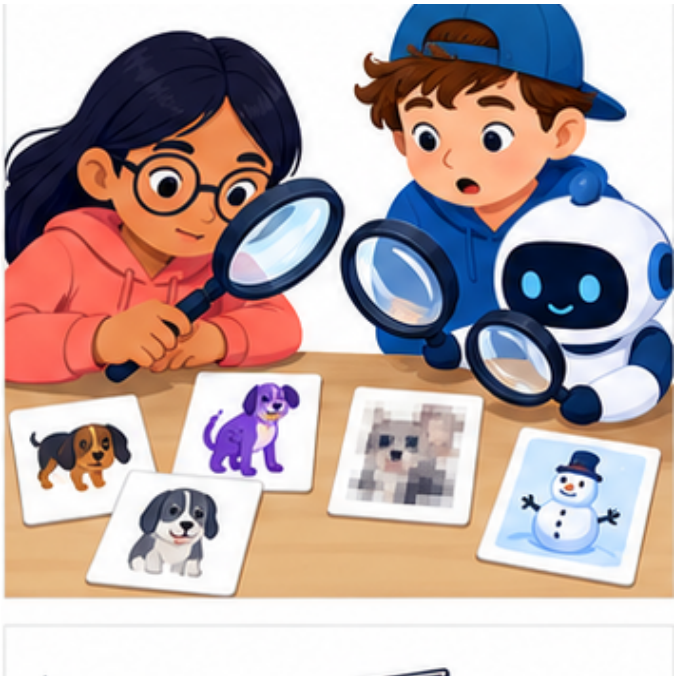
By the end, I can calculate mean and median and investigate an unusual value before changing it.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use noise and outliers to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

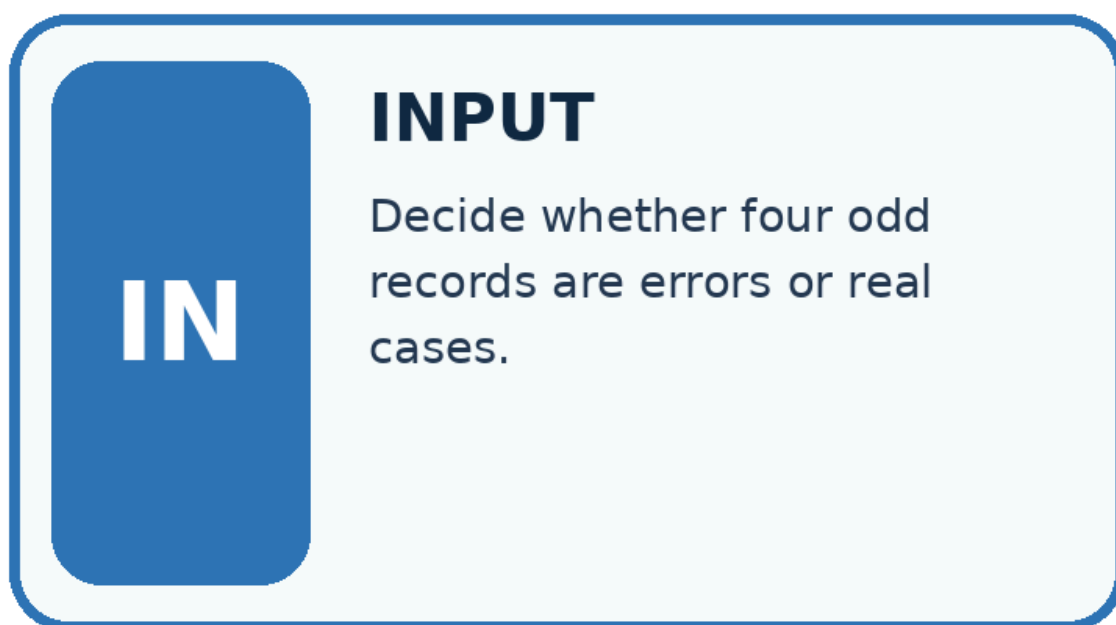
Five fictional activity durations are 2, 3, 3, 4 and 18 minutes. Most are near 3, but one is much longer. Different summaries reveal different aspects. Keep all values while investigating whether 18 is a genuine long activity or a recording error.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



PROCESS

Mistakes and unusual values should be investigated, not automatically erased.



OUT

OUTPUT

Decide whether four odd records are errors or real cases.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
values = [2, 3, 3, 4, 18]

ordered = sorted(values)

mean = sum(values) / len(values)

median = ordered[len(ordered) // 2]

print(mean)

print(median)
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: Sorted data

Evidence: 2, 3, 3, 4, 18

Meaning: Middle is 3

Step or case: Mean

Evidence: 30 divided by 5

Meaning: 6.0

Step or case: Typical middle

Evidence: Third of five values

Meaning: 3

Read each row from left to right and connect the evidence to the stated rule.

The script prints 6.0 and 3. `sorted` returns a sorted list without changing the original. This median expression works for a nonempty odd-length list only. For four sorted values, average the second and third. Do not use this odd-length shortcut for every dataset.

Use two summaries for two questions

The mean shares the total equally: $2 + 3 + 3 + 4 + 18$ is 30, and 30 divided by five is 6. The median asks

for the middle value after sorting: the third value is 3. The long duration 18 pulls the mean upwards but leaves the middle value at 3. A difference between the summaries is a clue to inspect the data, not permission to remove the inconvenient value.

Connect the picture and the explanation

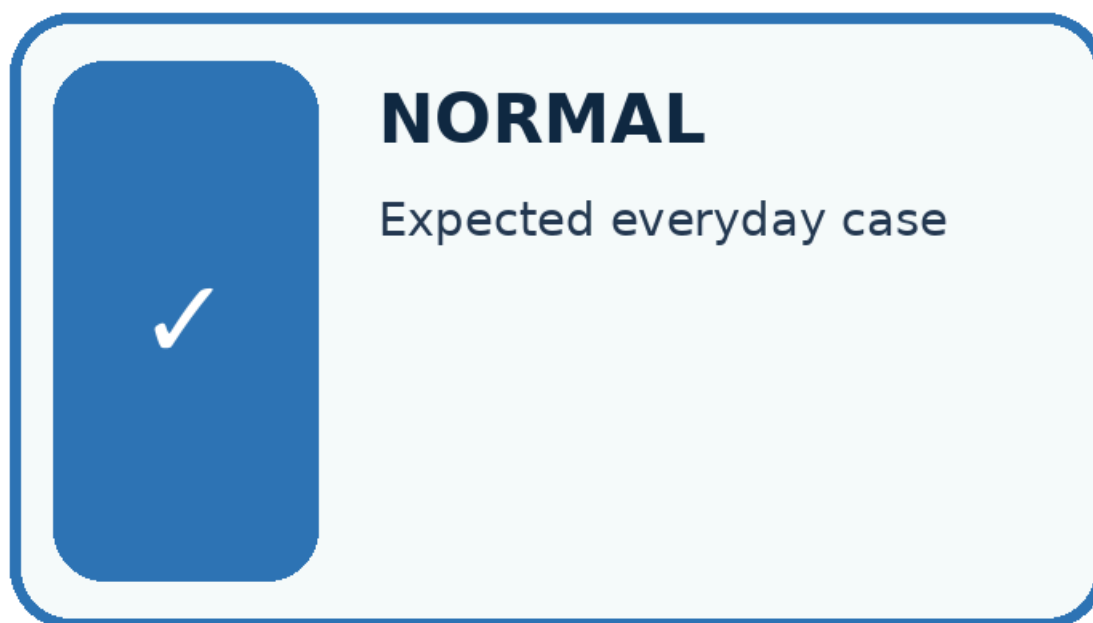
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

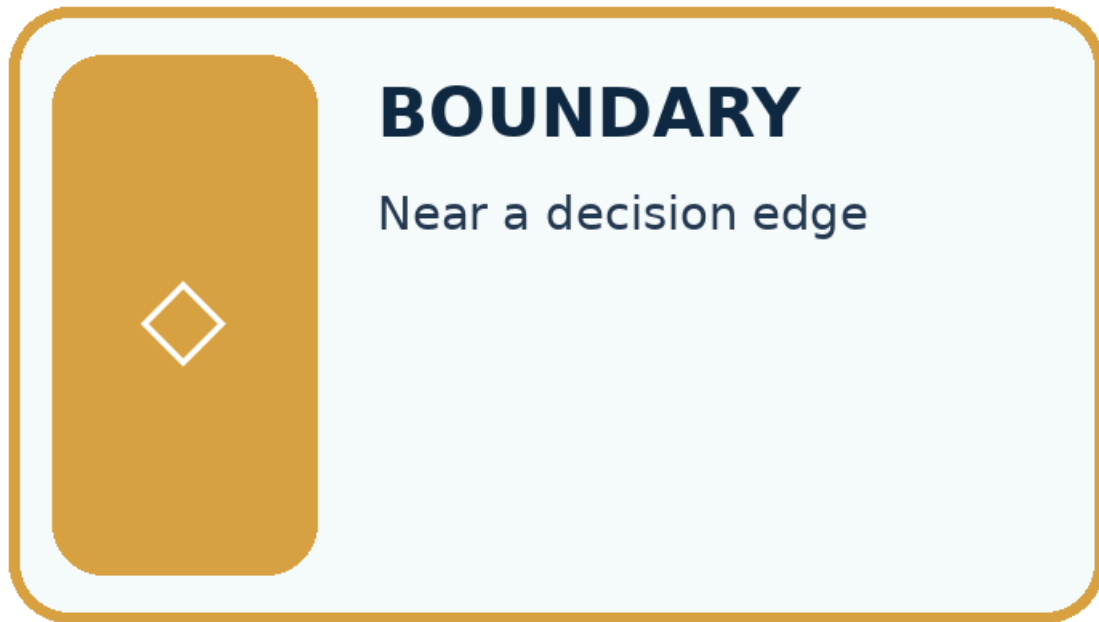
4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.

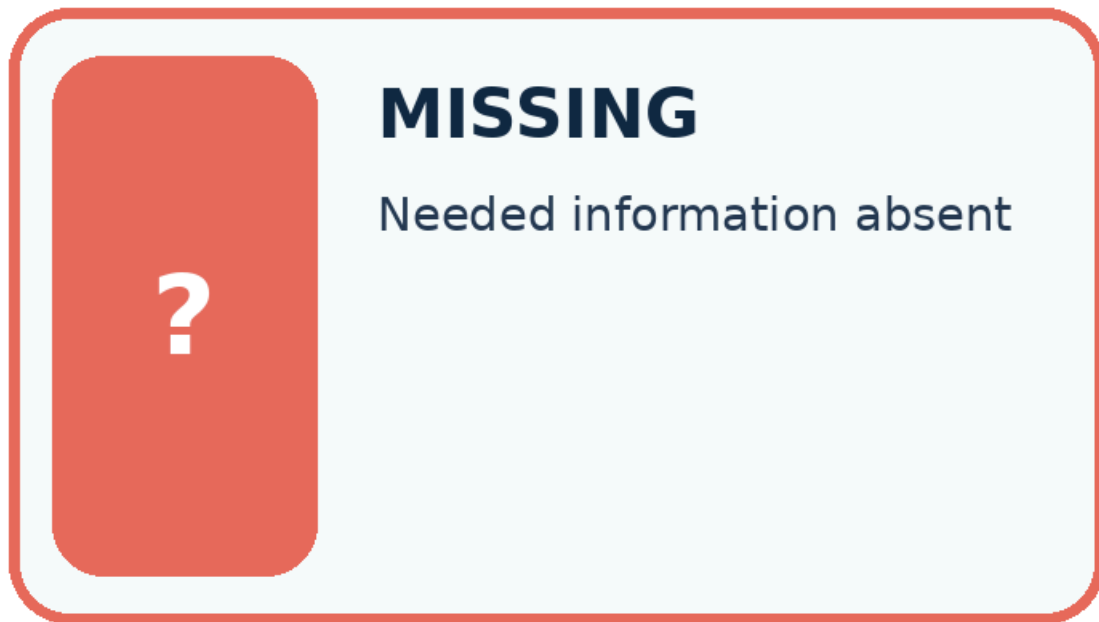




BOUNDARY

Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: An authorised log confirms 18 was a transcription error for 3.

Prediction: Corrected mean is 3 minutes and median is 3.

Reason: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

NOISE

incorrect or irrelevant
variation

2

OUTLIER

value far from most others

3

DUPLICATE

repeated record

4

INVESTIGATE

seek evidence before
changing data

Word · My own example · Not this / boundary

Word: noise

My own example: Variation or interference that obscures the information relevant to a task.

Not this / boundary: _____

Word: outlier

My own example: An observation far from a typical pattern under a stated comparison; it is not

automatically an error.

Not this / boundary: _____

Word: duplicate

My own example: A repeated record or copy; copies are not independent new observations.

Not this / boundary: _____

Word: investigate

My own example: Collect and examine evidence to answer a question rather than assuming a cause.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Mean

Meaning: Sum of numeric values divided by their count

Word: Median

Meaning: The middle of sorted values or mean of the two middle values

Word: Unusual value

Meaning: A value far from most others in a dataset

Word: Variation

Meaning: Differences among recorded values

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 17 • WORKED EVIDENCE

- 1 Most books: 40–300 pages; one record: 3,200 pages.
- 2 Check the source: extra zero, box set, or real long work?
- 3 Repair confirmed error; keep real rare case and document it.
- 4 Retest to learn whether the system handles legitimate extremes.

1. Most books: 40–300 pages; one record: 3,200 pages.
2. Check the source: extra zero, box set, or real long work?
3. Repair confirmed error; keep real rare case and document it.
4. Retest to learn whether the system handles legitimate extremes.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Replace 18 with 8. Calculate the new mean and median.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Noise and Outliers

A fictional library records five book-processing times in minutes.

The facts we will use

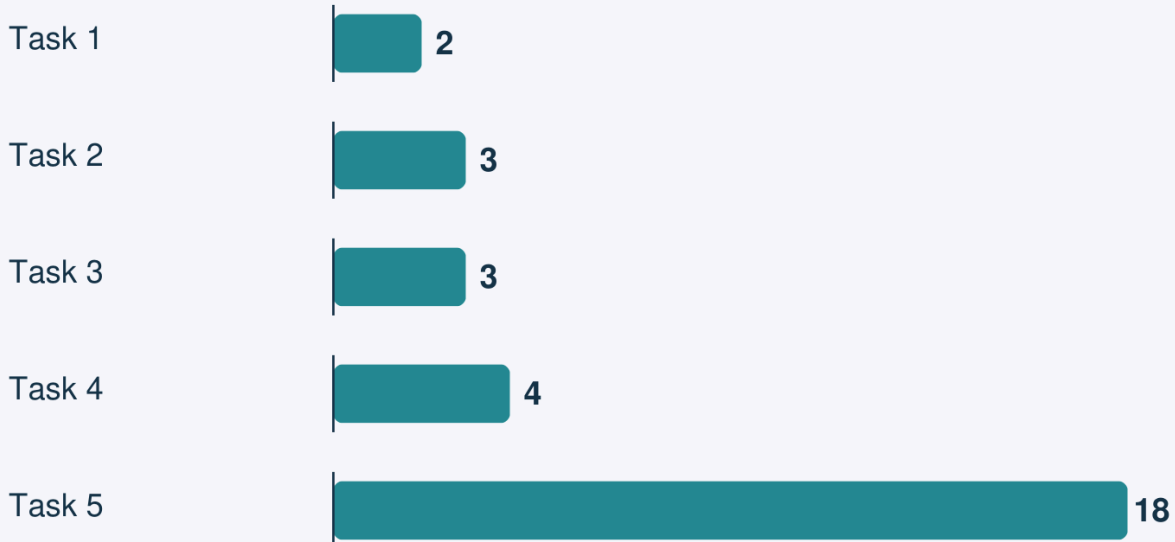
Sorted times: 2,3,3,4,18.

All are recorded measurements; no evidence yet shows 18 is a mistake.

AI ACADEMY / AI-2/17

Retain unusual recorded measurements

minutes



Bar length shows the supplied value; the marked baseline is zero.

Noise and Outliers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Mean

Sum 30; divide by 5: mean 6 minutes.



2. Median

The middle of five sorted values is 3 minutes.



3. Investigate 18

Check the source record and whether an unusually slow task occurred.



4. Document repair

Change a value only if evidence supports a correction; preserve the raw record.

CHANGE ONE THING

An authorised log confirms 18 was a transcription error for 3.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Mean

What happens and why: Sum 30; divide by 5: mean 6 minutes.

Step: 2. Median

What happens and why: The middle of five sorted values is 3 minutes.

Step: 3. Investigate 18

What happens and why: Check the source record and whether an unusually slow task occurred.

Step: 4. Document repair

What happens and why: Change a value only if evidence supports a correction; preserve the raw record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The mean uses every value and is pulled upward by 18. The median identifies the middle recorded time and stays at 3. Neither statistic declares the unusual observation invalid. Deleting it simply because it changes the mean would hide potentially real slow work and change the question being answered.

What this example does not establish

Five task times do not estimate all library processing times reliably or prove which summary suits every decision.

Read the labelled picture: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

Change one thing: An authorised log confirms 18 was a transcription error for 3. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Every unusual value is bad data that should be deleted. Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Removing 18 merely because it raises the mean would hide evidence. Check collection notes, keep an audit trail and compare summaries with and without it only with a clear explanation.

A dataset is empty. Is reporting a mean of 0 justified?

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Taking the middle item of an unsorted list can give the wrong median. For [18, 2, 4, 3, 3], the middle item is 4, but the sorted median is 3. Sort first. For an empty list, there is no mean or median under this lesson's rules.

Practise with less help: Compute two medians

Use workbook W2 for the connected example “Mean median and unusual values”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Mean median and unusual values. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Audit four suspicious records using keep / correct / remove / ask labels. Require a reason and a source check for every action.

Audit four suspicious records using keep / correct / remove / ask labels. Require a reason and a source check for every action.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER

1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

- 1 • NAME: Define the concept without repeating the textbook.
 - 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
 - 3 • APPLY: Solve a different example independently.
 - 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
 - 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

Keep legitimate unusual books so the recommender can serve unusual readers.

THIS WEEK'S CHECKPOINT Keep legitimate unusual books so the recommender can serve unusual readers.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Investigate unusual recommender features and show evidence-backed repairs instead of removing inconvenient records.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Decide whether four odd records are errors or real cases.

CHALLENGE Decide whether four odd records are errors or real cases.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 17 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Recorded lengths are 90,95,100,105,900 pages.

The question

Is 900 automatically wrong?

Worked explanation

No. It is an outlier, but may be a real large book. Check catalogue, units and entry process.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

An outlier needs investigation

Evidence or stage · Value or result

Evidence or stage: Book A

Value or result: 90

Evidence or stage: Book B

Value or result: 95

Evidence or stage: Book C

Value or result: 100

Evidence or stage: Book D

Value or result: 105

Evidence or stage: Book E

Value or result: 900

Recorded lengths are 90,95,100,105,900 pages.

Is 900 automatically wrong?

No. It is an outlier, but may be a real large book. Check catalogue, units and entry process.

14. Try a new case independently

Independent case: Compare summaries independently

For 1, 2, 2, 3 and 12, calculate mean and median. Then replace 12 with 2 and recalculate. Explain the change.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 17 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Page counts are 40, 42, 43 and 900. Should 900 be deleted immediately? Give two explanations to investigate.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Keep legitimate unusual books so the recommender can serve unusual readers.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Page counts are 72,80,96,960. The last book may be an anthology. Should a cleaning script delete it automatically? Give a defensible investigation and document the outcome.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a

model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional library records five book-processing times in minutes.

By the end, I can · How I will show it

By the end, I can: Calculate mean and median

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Investigate an unusual value before deleting it

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Explain how a summary can hide variation

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Calculate mean and median

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional library records five book-processing times in minutes.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Change a value only if evidence supports a correction; preserve the raw record.

1. Median
2. Investigate 18
3. Document repair
4. Mean

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Document repair Change a value only if evidence supports a correction; preserve the raw record. The mean uses every value and is pulled upward by 18. The median identifies the middle recorded time and stays at 3. Neither statistic declares the unusual observation invalid. Deleting it simply because it changes the mean would hide potentially real slow work and change the question being answered.

Practice 2: explain the deciding evidence

Changed-case practice: An authorised log confirms 18 was a transcription error for 3. Which conclusion is supported by the supplied facts?

1. Corrected mean is 3 minutes and median is 3.
2. We can decide the result without examining the changed input or the procedure.

3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Corrected mean is 3 minutes and median is 3. Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Repair a missing building block

Review Overfitting Intuition (ai-2/16). Goal: Compare training and validation evidence for model capacity. Short practice: Does zero training error justify choosing Complex in the original record?. Return to this week and explain what you changed.

Review Decision Trees (ai-2/14). Goal: Fit a root question and leaf predictions from training examples. Short practice: Why does changing colour alone not change this tree's prediction?. Return to this week and explain what you changed.

Review Training and Testing (ai-2/11). Goal: Choose a split unit that matches the intended prediction claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan counts?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.